

Yap Hong Ping

Singapore PR | +65-97773450 | yhp2130@gmail.com | linkedin.com/in/yaphongping | yhp2130.github.io
Agentic AI | RAG | NLP | Computer Vision | MLOps

1 PROFESSIONAL SUMMARY

Applied AI Engineer and Data Scientist with 8+ years delivering end-to-end production AI systems — Agentic RAG pipelines, LLM fine-tuning (BART, RoBERTa), computer vision, and full-stack deployment (FastAPI, React.js, OpenShift CI/CD | cloud-compatible architecture (on-prem and cloud-portable)). Delivered measurable outcomes across 5 shipped systems: 96% reduction in knowledge-retrieval time (2hr to 5min), F1=0.8 NLP clause extraction (200 contracts), 70% CV accuracy (receipt OCR), 100+ invoices automated/month, and concept-to-production MVPs in under 2 weeks using GitHub Copilot SpecKit. Open to Applied AI Engineer and Data Scientist roles in software, fintech, and deep-tech industries

2 CORE COMPETENCIES

- Data Science: Frames ambiguous problems into measurable AI objectives; iterates via experimentation, RAGAS evaluation, and error analysis
- AI Application: Delivers ML, Deep Learning (CV & NLP), and Generative AI (LLM/RAG) with user requirements, feedback, and adoption in scope
- Full-Stack Deployment: Integrates models into web applications (React.js, Vue.js, FastAPI) meeting enterprise architecture standards
- Stakeholder Engagement: Agile delivery with regular demos, retrospectives, and change management driving measurable production adoption

3 TECHNICAL SKILLS

- ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, HuggingFace (BART, RoBERTa, Nougat, DETR, LayoutLM)
- Agentic AI & RAG: LangChain (ReAct, tool-calling), OpenAI AsyncOpenAI, Elasticsearch kNN | HNSW vector database, Agentic Chunking, Prompt Engineering
- Backend & Full-Stack: FastAPI, FastMCP, React.js (TypeScript/SSE), Vue.js, Streamlit, Docker, Kubernetes, OpenShift
- MLOps: GitLab CI/CD, ArgoCD, Helm, Artifactory, OpenShift (dev/staging/prod) | experiment tracking | model monitoring
- Data: Elasticsearch, RDBMS (Oracle SQL, MySQL, SQLite), Logstash | vector DB (Elasticsearch kNN/HNSW)
- AI Platforms: Datarobot, Dataiku, Dify, SAP Joule, Copilot Studio, GitHub Copilot

4 PROFESSIONAL EXPERIENCE

4.1 INFINEON TECHNOLOGIES ASIA PACIFIC — STAFF ANALYST, DATA SCIENTIST (NOV 2021–PRESENT)

Agentic Process Knowledge RAG

- Delivered production-ready multi-agent virtual consultant in under 2 weeks (GitHub Copilot SpecKit); custom asyncio ReAct — Planner to Phase Sub-Agents (5 specialisations) to Checker; ExecutionPlan/SubGoal dependency graph; 4 knowledge sources (ARIS, Confluence, Intranet, PRIMA)
- React.js + TypeScript frontend with SSE streaming; per-message thumbs-up/down feedback (SQLite); GitLab CI/CD to ArgoCD to Helm to OpenShift (dev/staging/prod) + Artifactory; per-agent YAML config
- **Stack:** Python | OpenAI AsyncOpenAI | FastAPI | React.js | GitLab CI/CD | ArgoCD | Helm | OpenShift | Artifactory

Document Retrieval & Agentic RAG Chatbot

- 3-generation evolution across approx. 1,000 documents (TF-IDF+SBERT to Elasticsearch kNN to LangChain ReAct agent); 96% query time reduction (2hr to 5min); 5 chunking strategies evaluated — agentic chosen; SFT-Mistral embedder via RAGAS; Docling + EasyOCR ingestion

- 3 role-gated LangChain tools (abbreviation, kNN search, BART contract clauses via LLM-to-SQL); approx. 100 queries/day; QnA feedback loop for continuous improvement
- **Stack:** Python | LangChain | Elasticsearch (kNN HNSW) | SFT-Mistral | Docling | FastAPI | React.js | Nginx | OpenShift

NLP Contract Clause Extraction

- Fine-tuned RoBERTa-large-MNLI (8-class logistics classifier, F1=0.8) + BART seq2seq on 200 labelled contracts; Nougat PDF transcription + DETR table detection preprocessing; human-in-the-loop Vue.js review; wrapped as LangChain agent tool inside ClaireGPT
- **Stack:** Python | HuggingFace (RoBERTa-MNLI, BART) | PyTorch | Nougat | DETR | FastAPI | Vue.js | GitLab CI/CD | Oracle DB

Travel Expense Receipt Intelligence

- SqueezeNet CNN (7 doc classes) + Faster RCNN segmentation + EasyOCR (custom weights) + regex NER; 7-flag rule validator; event-driven SAP Concur v4 webhook replacing batch; 70% accuracy on test samples
- **Stack:** Python | PyTorch (SqueezeNet | Faster RCNN) | EasyOCR | FastAPI | SAP Concur v4 API | SAP BTP

Distributor Invoice Audit Automation

- Configurable rule engine for AP invoice reconciliation (100+ invoices/month): 6 match flags (SMD, qty, price, currency, SAP#, date); fuzzy matching (thefuzz) + LLM fallback for CJK company names; ABBYY Vantage multilingual extraction (Chinese, German, English); Vue.js exception dashboard
- **Stack:** Python | ABBYY Vantage | FastAPI | Vue.js | Elasticsearch | Logstash | Docker
- **Team & Delivery:** Agile 2-week sprints, stakeholder demos and retrospectives; intern mentoring with structured onboarding and task reviews

4.2 PSA CORPORATION — PRINCIPAL ENGINEER, SMART SYSTEMS SOLUTIONS (JUL 2018–NOV 2021)

- Predictive Maintenance (Maritime Equipment): Semi-supervised pipeline (anomaly detection to supervised classification) on sensor/log data from over 20 machines at 0.5s sampling; approx. 8 man-hours/week saved via continuous monitoring dashboard | Stack: Python | Scikit-learn
- CV Equipment Monitoring (POC): Object detection for structural crack and roller wear defect detection; custom training with image augmentation | Stack: Python | PyTorch
- Cloudera Hadoop data pipelines for large-scale industrial sensor ingestion and feature engineering; integrated with real-time monitoring dashboards | Stack: Python | Hadoop | Spark

4.3 PSA CORPORATION — SENIOR MECHANICAL ENGINEER (JUL 2013–JUL 2018)

- Automated component life tracking and failure pattern analysis; translated mechanical domain expertise into predictive monitoring criteria applied in subsequent ML system design

5 EDUCATION

- Master of Technology (Knowledge Engineering), National University of Singapore – Institute of Systems Science (NUS-ISS) | 2018–2020
- Bachelor of Engineering (Mechatronic Engineering), Nanyang Technological University (NTU) | 2009–2013